

Career Switch Prediction Using Machine Learning

Ahsanul Fahim Ahmed
Department of Computer Science and Engineering
BRAC University
ahsanul.fahim.ahmed@g.bracu.ac.bd

Abstract—Career switching has become increasingly common due to rapid technological changes, evolving job requirements, and the need for continuous upskilling. This paper presents a machine learning-based approach for predicting whether a professional is likely to change their career using demographic, educational, and work-related attributes. Multiple supervised learning models and one unsupervised learning technique are applied to a real-world dataset with class imbalance. Experimental results show that the Decision Tree classifier performs best overall in terms of accuracy, F1-score, and ROC-AUC, demonstrating strong performance under non-linear feature interactions.

Index Terms—career switch prediction, machine learning, classification, imbalanced dataset, model comparison

I. PROJECT RESOURCES

The source code, dataset, and all experimental artifacts associated with this project are publicly available in the accompanying GitHub repository.

II. INTRODUCTION

The objective of this project is to predict whether a professional is likely to change their career using machine learning techniques. Career switching has become increasingly common due to rapid technological changes, evolving job requirements, and the need for continuous upskilling.

This project aims to solve a real-world prediction problem by analyzing demographic information, education level, work experience, company-related attributes, and training data. By applying multiple supervised learning models and one unsupervised learning technique, the project evaluates model performance and identifies the most suitable approach for career switch prediction.

The motivation behind this project is to demonstrate the application of machine learning concepts such as preprocessing, exploratory data analysis, classification, clustering, and model evaluation on a realistic dataset.

III. DATASET DESCRIPTION

A. Dataset Overview

- **Number of data points:** 5000
- **Number of features:** 14 (raw), 168 after encoding
- **Problem type:** Classification

This is a classification problem because the target variable `will_change_career` has discrete class labels (0 and 1), rather than continuous numerical values.

B. Feature Types

The dataset contains:

- **Quantitative features:** training hours, city development index, experience-related variables
- **Categorical features:** city, education level, company type, last job, etc.

C. Encoding of Categorical Variables

Categorical variables must be encoded because machine learning models require numerical input. One-hot encoding was applied to convert categorical features into binary numeric columns without introducing artificial ordering.

D. Correlation Analysis

A correlation heatmap was generated using the Seaborn library to analyze relationships between features and the target variable.

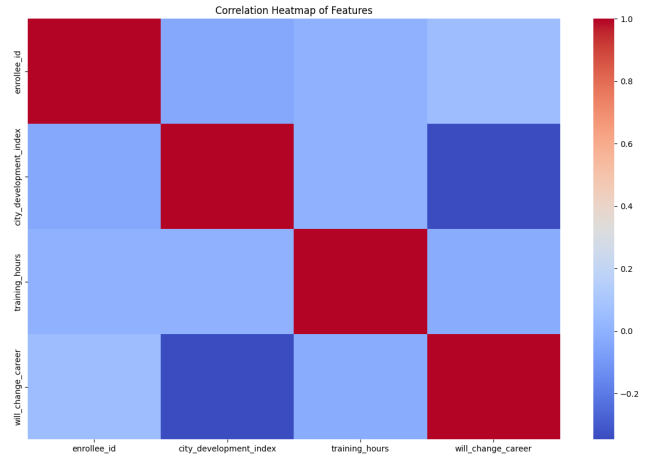


Fig. 1: Correlation heatmap of numerical features.

E. Correlation Interpretation

The correlation analysis shows that no single feature has a very strong correlation with career switching. Instead, career switching behavior depends on multiple interacting features, indicating a non-linear problem structure.

F. Imbalanced Dataset

The output feature `will_change_career` does not have equal instances:

- Class 0 (No career switch): 3738
- Class 1 (Career switch): 1262

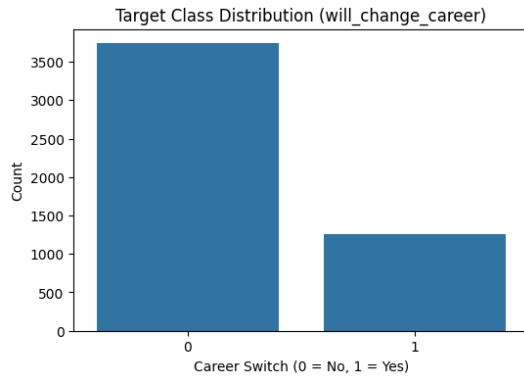


Fig. 2: Class distribution for will_change_career.

Since the dataset is imbalanced, evaluation metrics such as precision, recall, F1-score, and AUC are more informative than accuracy alone.

G. Exploratory Data Analysis (EDA)

Exploratory Data Analysis revealed several important relationships:

- Higher training hours are associated with higher career switching tendency
- City development index influences switching behavior
- Company-related attributes play a significant role

IV. DATASET PRE-PROCESSING

A. Missing Values

Cause: Some features contain missing values which machine learning models cannot process.

Solution: Mean imputation for numerical features and mode imputation for categorical features.

Listing 1: Missing value handling.

```
1 SimpleImputer(strategy='mean')
2 SimpleImputer(strategy='most_frequent')
```

B. Categorical Values

Cause: Categorical features cannot be used directly by ML models.

Solution: One-hot encoding was applied.

C. Feature Scaling

Cause: Distance-based and neural network models are sensitive to feature scale.

Solution: StandardScaler was applied after train-test split to prevent data leakage.

D. Identifier Column

Cause: enrollee_id is an identifier and does not contribute to prediction.

Solution: The ID column was removed.

V. DATASET SPLITTING

The dataset was split using stratified sampling:

- Training set: 80%
- Test set: 20%

Stratification ensures class distribution is preserved in both sets.

VI. MODEL TRAINING AND TESTING

A. Supervised Learning Models

The following supervised models were trained:

- K-Nearest Neighbors (KNN)
- Decision Tree
- Neural Network (MLP)

B. Unsupervised Learning: K-Means

The problem was also treated as an unsupervised learning task using K-Means clustering to identify hidden patterns.

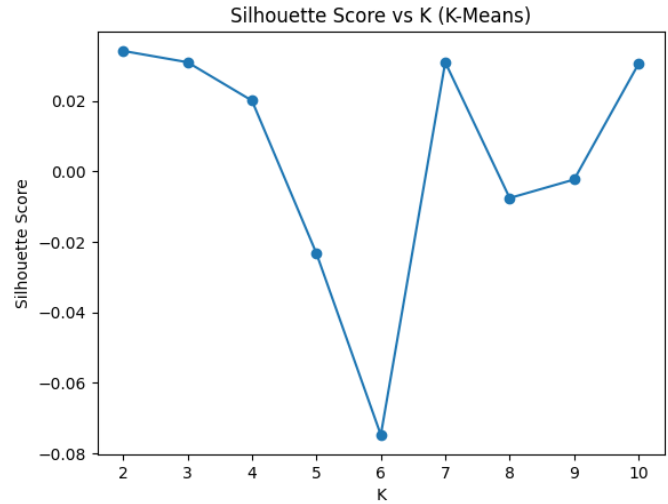


Fig. 3: Silhouette score vs. number of clusters (K-Means).

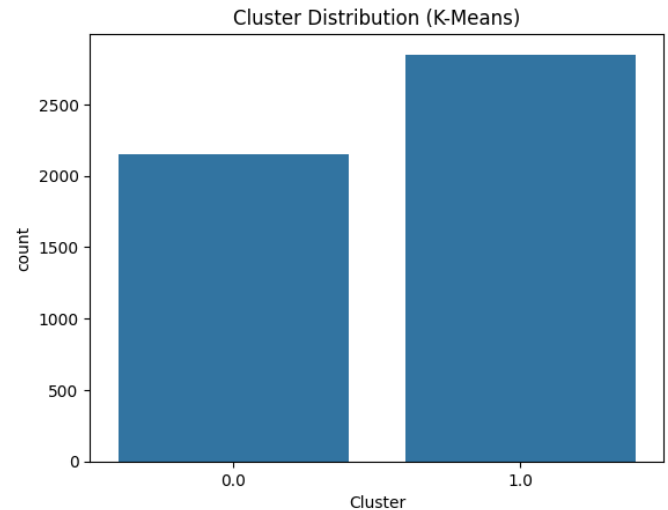


Fig. 4: Cluster distribution (K-Means).

VII. MODEL SELECTION AND COMPARISON

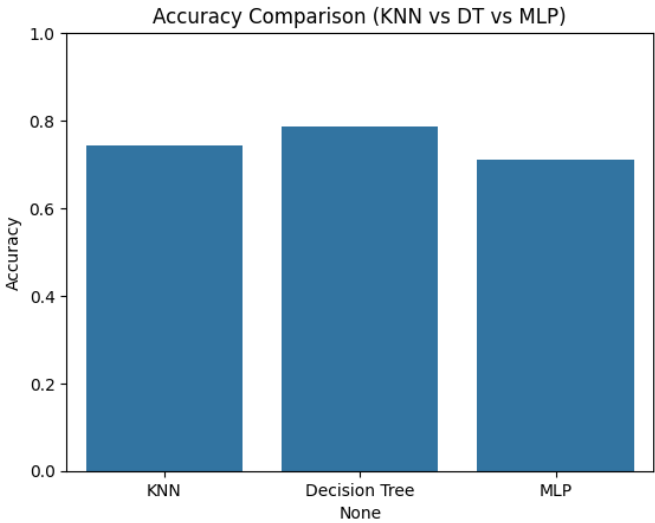


Fig. 5: Accuracy comparison of supervised models (KNN vs. Decision Tree vs. MLP).

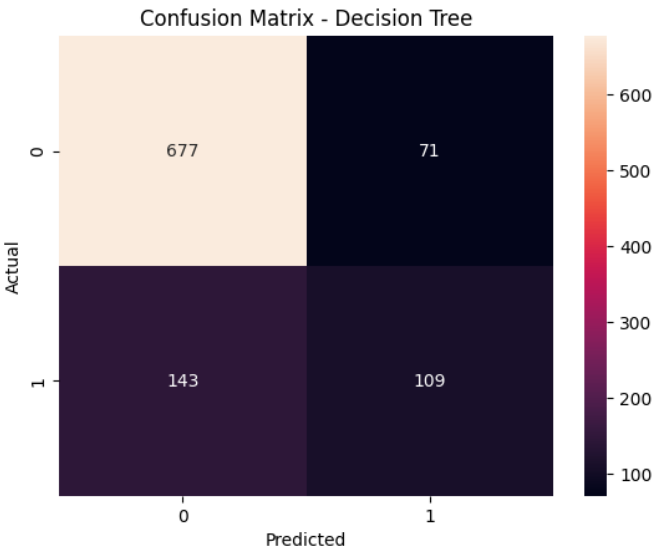


Fig. 7: Confusion matrix for Decision Tree.

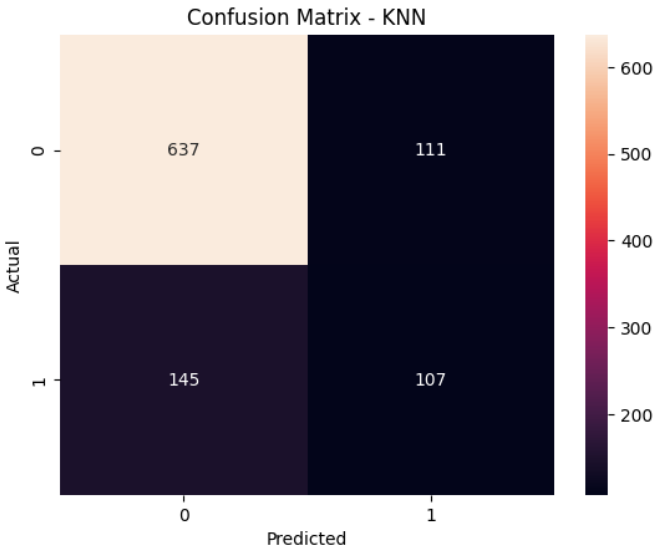


Fig. 6: Confusion matrix for KNN.

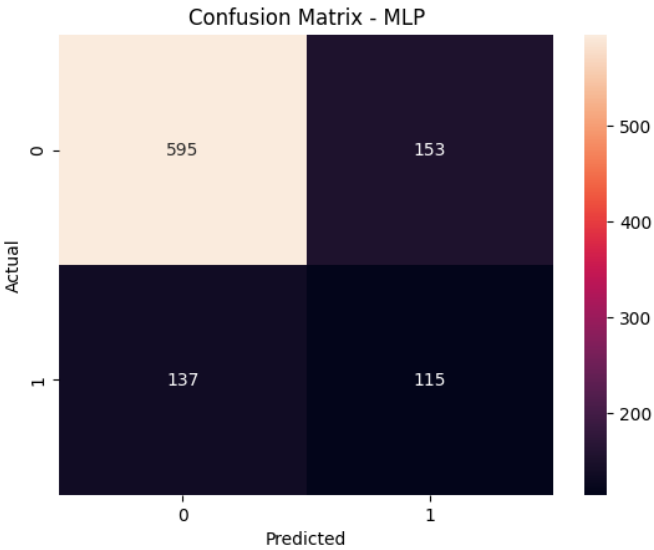


Fig. 8: Confusion matrix for MLP.

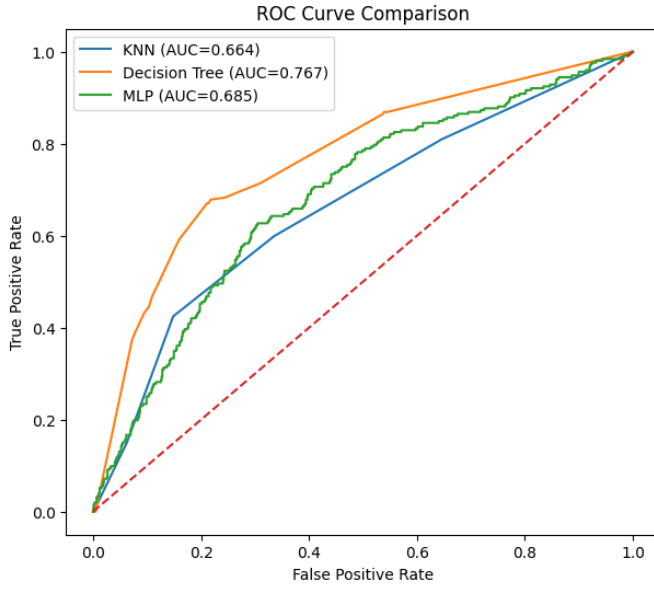


Fig. 9: ROC curve comparison of supervised models.

TABLE I: Model comparison table.

Model	Accuracy	Precision	Recall	F1	AUC
KNN	0.744	0.491	0.425	0.455	0.664
Decision Tree	0.786	0.606	0.433	0.505	0.767
MLP	0.710	0.429	0.456	0.442	0.685

VIII. CONCLUSION

From the results, it is evident that the Decision Tree model performed best overall. It achieved the highest accuracy, F1-score, and AUC, indicating a balanced and reliable performance.

The Decision Tree performs well because it can model non-linear relationships and feature interactions effectively without extensive tuning. KNN served as a baseline model, while the Neural Network showed higher recall but lower precision.

A. Challenges Faced

- Class imbalance
- High dimensionality after encoding
- Sensitivity of models to scaling
- Preventing data leakage